PYTHON BASICS



DIFFERENT IMPLEMENTATIONS OF PYTHON

Implementation	Virtual Machine	Ex) Compatible Language
CPython	CPython VM	C
Jython	JVM	Java
IronPython	CLR	C#
Brython	Javascript engine (e.g., V8)	JavaScript
RubyPython	Ruby VM	Ruby

toptal

DATA TYPES:

- **Primitive Data Structures** – These data structures contain simplified data values and serve as the foundation for manipulating data. The four primitive data structures are *integers*, *float*, *string*, and *boolean*.
- **Non-primitive Data Structures** – These data structures store values, as well as a collection of values, in varying formats. The four built-in non-primitive data structures are *lists*, *tuples*, *dictionaries*, and *sets*.

PRIMITIVE:

1. Integers in Python

In Python, integers are zero, positive, or negative whole numerals. They do not have a fractional part and have unlimited precision.

Integers can have octal, binary, or hexadecimal values.

2. Float in Python

In Python, float represents the floating point number. They're used to represent real numbers and are written with a decimal point that divides the integer and fractional parts.

PRIMITIVE:

3. String in Python

In Python, a string is a collection of alphabets, characters, or words. While it's one of the most primitive of data structures, it's also vital to data manipulation. Python has a built-in string class known as str, which is immutable.

4. Boolean in Python

This is one of Python's built-in data types, and it represents the truth value of an expression. The Python boolean type has only two possible values: true or false. In other words, you can evaluate any expression and get either true or false in response.

NON PRIMITIVE:

1. Lists in Python

- A list is an ordered collection of items and is one of the most essential data structures to implement in any Python project. As these are "ordered collections," each item in the list has an order which uniquely identifies it.
- You can assign addresses to each element of the list (which are indexes). The index value begins at zero and continues on until the last element (the positive index). There is also the negative index, which starts at -1. This enables you to access elements from the very end to the very beginning.
- When creating a list, you should enclose all the items on the list within square brackets, and separate them by commas.
- **List_C = [item 1, item 2, item 3....., item n]**
- Lists created in Python are mutable—meaning you can modify them after you've created them. This allows a user to search, add, shift, delete, and move elements from a list.